POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2022
Programme: BE | Full Marks: 100
Course: Discrete Structure | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Mention the types of set operations that you are familiar with. In a class of 40 students, 20 have chosen Discrete Mathematics but not Calculus. If every student has chosen either Discrete Mathematics or Calculus or both, find the number of students who chose both Discrete Mathematics and Calculus and the number of students to choose Calculus but not Discrete Mathematics. 8

   b) How do you differentiate Euclid's Division Algorithm and Extended Euclidean Algorithm? Find the gcd (83, 19) by Euclid's Division Algorithm. 7

OR

Give the type of Boolean Functions. Prove that "if n is a composite integer, then n has a prime divisor less than or equal to square root of n.

2. a) Define closure of a relation. Let A= {1, 2, 3, 4} and let R = {(1, 2), (2, 3), (3, 4), (2, 1)}. Find the transitive closure of R using Warshall's algorithm. 8

   b) Explain the closure of relation. Let R be the relation of set {0,1,2,3} containing order pair (0,1),(1,1),(1,2),(2,0),(2,2) and (3,0). Find the symmetric closure of R. 7

OR

Define POSET.Show that the relation R= {(a, b)}| a divides b} defined on a set S = {1, 2, 3, 4, 6} is a Partial order relation.

3. a) Find the explicit formula for the Fibonacci Series, $f_n = f_{n-1} + f_{n-2}$ with $f_0=0$ and $f_1=1$. 7

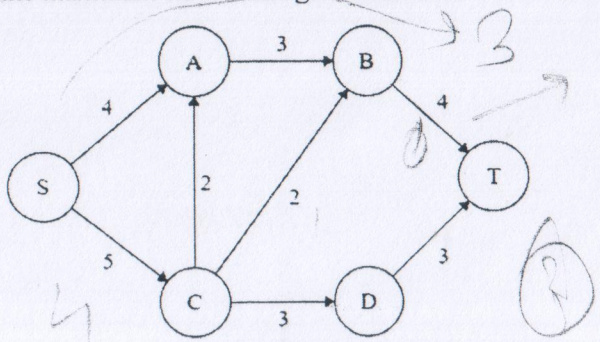
   b) Solve the recurrence relation: $\mathbf{a_n - 6a_{n-1} + 8a_{n-2} = 3}$; $a_0 = 10$, $a_1 = 25$. 8

4. a) Define the terms Tautology, Contradiction and Logical Equivalences. Show that Implication and Contrapositive are Logically Equivalent. State the converse, contrapositive and Inverse of the statement: "A positive integer is prime only if it has no divisors other than 1 and itself. 7

   b) Prove the validity of the following statements, "It is not sunny this afternoon and it is colder than yesterday". "We will go swimming only if it is sunny". "If we don't go swimming, then we will go for the canoe trip". "If we go for the canoe trip, then we will be home by sunset" , Which leads to the conclusion "We will be home by sunset". 8

5. a) Use Mathematical induction to show that: $\mathbf{8^n - 3^n}$ is divisible by 5. 7

   b) Find the maximum flow in the given network: 8



6. a) What do you understand by the walk, path, trail and circuits in the graph theory? Explain briefly with figures. 8

   b) What are difference between the DFA and NFA? Construct the DFA that accepts any strings over {a, b} that doesn't contain the string 'aabb' in it. 7

7. Write short notes on: (**Any two**) 2×5

   a) Regular Expressions

   b) Graph representation techniques

   c) TOH puzzle